

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test Case 1

Investigate the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a transmission line using the following primary constants: a) Resistance, $R = 10 \Omega/\text{km}$

b) Capacitance, $C = 8 \mu\text{F}/\text{km}$

c) Inductance, $L = 3.5 \text{ mH}/\text{km}$

d) Conductance, $G = 4 \mu\text{mho}/\text{km}$

e) Frequency, $f = 2 \text{ kHz}$

Scilab Program

```
clear; clc; R = 10; L = 3.5e-3; f = 2*10^3; G = 4*10^-6;
C = 8*10^-6; w = 2*pi*f; Z = R + (%i*w*L);
Y = G + (%i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot displays the Scilab 2026.1.0 environment. The left pane shows the Scilab Console with the following output:

```
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

The right pane shows the Scilab Editor with the source code of the program, which is identical to the code block provided above. The editor window title is 'exp 1.1.sci (C:\Users\sai\transmission line\exp 1.1.sci) - SciNotes'.

Result Table

S.No	Parameters	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	R=10, L=3.5mH, C=8μF, G=4μmho, f=2kHz	21.050 - j2.362	0.2376	2.1161	0.2376	2.1161

Test Case 2

Examine the impact of resistance and inductance on Z and γ at f = 2.5 kHz. Vary R from 10 Ω to 50 Ω in steps of 10 Ω (L, G, C held constant); then vary L from 3.5 mH to 11.5 mH in steps of 2.5 mH (R, G, C held constant).

Scilab Program (update R or L per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set R
(or L) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot shows the Scilab environment. On the left, the 'Console' window displays the output of the program for varying R from 10 to 50 Ohms. On the right, the 'Scilab Editor' shows the code used to generate this output.

Console Output:

S.No	R	Zo (ohms)	Alpha (Np/km)	Beta (rad/km)	Alpha (The)	Beta (The)
1	10	14.85+j-1.34	0.33671	3.73240	0.33671	3.73240
2	20	15.03+j-2.65	0.66557	3.77629	0.66557	3.77629
3	30	15.30+j-3.90	0.98067	3.84435	0.98067	3.84435
4	40	15.64+j-5.09	1.27874	3.93096	1.27874	3.93096
5	50	16.04+j-6.20	1.55883	4.03078	1.55883	4.03078

Scilab Editor Code:

```
1 clear;
2 clc;
3
4 L = 3.5e-3;
5 G = 4e-6;
6 C = 16e-6;
7 f = 2.5e3;
8
9 w = 2*pi*f;
10
11 printf("\nVARIABLE-R\n");
12 printf("-----\n");
13 printf("S.No\tR\tZo (ohms)\t\t\t\tAlpha (Np/km)\t\tBeta (rad/km)\t\tAlpha (The)\t\tBeta (The)\n");
14 printf("-----\n");
15
16 k = 1;
17
18 for R = 10:10:50
19
20     Z = R + i*w*L;
21     Y = G + i*w*C;
22
23     Zo = sqrt(Z/Y);
24     gamma = sqrt(Z*Y);
25
26     alpha = real(gamma);
27     beta = imag(gamma);
28
29     printf("%d\t%d\t%.2f+j%.2f\t%.5f\t%.5f\t%.5f\t%.5f\n",...
30         k,R,real(Zo),imag(Zo),alpha,beta,alpha,beta);
31
32     k = k + 1;
33
34 end
```

Variable R (constants: $L = 3 \text{ mH}$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	R (Ω)	Simulation Results			Theoretical Results	
		Z (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	10	19.532 – j2.546	0.2560	1.9635	0.2560	1.9635
2	20	19.994 – j4.975	0.5002	2.0100	0.5002	2.0100
3	30	20.667 – j7.220	0.7258	2.0777	0.7258	2.0777
4	40	21.468 – j9.267	0.9316	2.1582	0.9316	2.1582
5	50	22.337 – j11.133	1.1192	2.2456	1.1192	2.2456

Variable L (constants: $R = 10 \Omega$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	L (mH)	Simulation Results			Theoretical Results	
		Z (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	3.5	21.050 – j2.363	0.2375	2.1161	0.2375	2.1161
2	6.0	27.446 – j1.812	0.1822	2.7592	0.1822	2.7592
3	8.5	32.632 – j1.524	0.1532	3.2805	0.1532	3.2805
4	11.0	37.105 – j1.340	0.1348	3.7302	0.1348	3.7302

Test Case 3

Explore the impact of frequency and capacitance on Z and γ . Vary f from 2 kHz to 10 kHz in steps of 2 kHz (L, R, G, C held constant); then vary C from 8 μF to 16 μF in steps of 2 μF (R, G, L held constant).

Scilab Program (update f or C per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set f
(or C) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the results of calculations for characteristic impedance Z_0 , attenuation constant a , and phase constant b . The editor on the right shows the corresponding MATLAB code in a file named 'exp 1.3.sci'.

```

Scilab 2026.1.0 Console
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->

exp 1.3.sci (C:\Users\sahatransmission line\exp 1.3.sci) - SciNotes
exp 1.1.sci X exp 2.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X

1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(i*w*L);
5 Y=G+(i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('Zo = %f + j(%f) ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 al=round(a*10000)/10000;
13 printf('Attenuation constant a = %f neper/km\n',al);
14 b=imag(P);
15 bl=round(b*10000)/10000;
16 printf('Phase constant b = %f radians/km\n',bl);
17

```

Variable f (constants: $R = 10 \Omega$, $L = 3 \text{ mH}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	f (Hz)	Simulation Results			Theoretical Results	
		Z_0 (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	2000	19.532 - j2.546	0.2560	1.9635	0.2560	1.9635
2	4000	19.407 - j1.281	0.2576	3.9021	0.2576	3.9021
3	6000	19.384 - j0.855	0.2580	5.8460	0.2580	5.8460
4	8000	19.376 - j0.642	0.2581	7.7914	0.2581	7.7914
5	10000	19.372 - j0.513	0.2581	9.7373	0.2581	9.7373

Variable C (constants: $R = 10 \Omega$, $L = 9 \text{ mH}$, $f = 2500 \text{ Hz}$, $G = 0.4 \times 10^{-6}$)

S.No	C (μF)	Simulation Results			Theoretical Results	
		Z_0 (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	8	33.562 - j1.185	0.1490	4.2175	0.1490	4.2175
2	10	30.019 - j1.060	0.1666	4.7153	0.1666	4.7153
3	12	27.403 - j0.968	0.1825	5.1654	0.1825	5.1654
4	14	25.370 - j0.896	0.1971	5.5793	0.1971	5.5793
5	16	23.732 - j0.838	0.2107	5.9645	0.2107	5.9645

Result

The characteristic impedance and propagation constant of the transmission line were estimated for varying R , L , f and C , and the simulation results (Scilab) were verified against the theoretical values calculated from $Z = R + j\omega L$ and $Y = G + j\omega C$.